

CoRoT-1b

R-2449

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_210217 · created 2026-08-22T23:53:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459262.7924 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.042
Transit depth (Rp/Rs)^2	2.04 +/- 1.19 [%]
Orbital Inclination (inc)	89.43 +/- 2.82
Airmass coefficient 1 (a1)	1.24 +/- 0.37
Airmass coefficient 2 (a2)	-0.122 +/- 0.054
Scatter in the residuals of the lightcurve fit is	28.64 %
Best Comparison Star	#3 - [486, 102]
Optimal Aperture	3.0
Optimal Annulus	6.51
Transit Duration (day)	0.1123 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.042 vs 0.1388 (deviation 0.08σ)
Duration fit vs published	0.1123 d vs 0.1046 d (deviation 1.07σ)
Mid-time O–C	10.9 min
Depth SNR	2.7
Residual scatter	28.64 %

Light curve

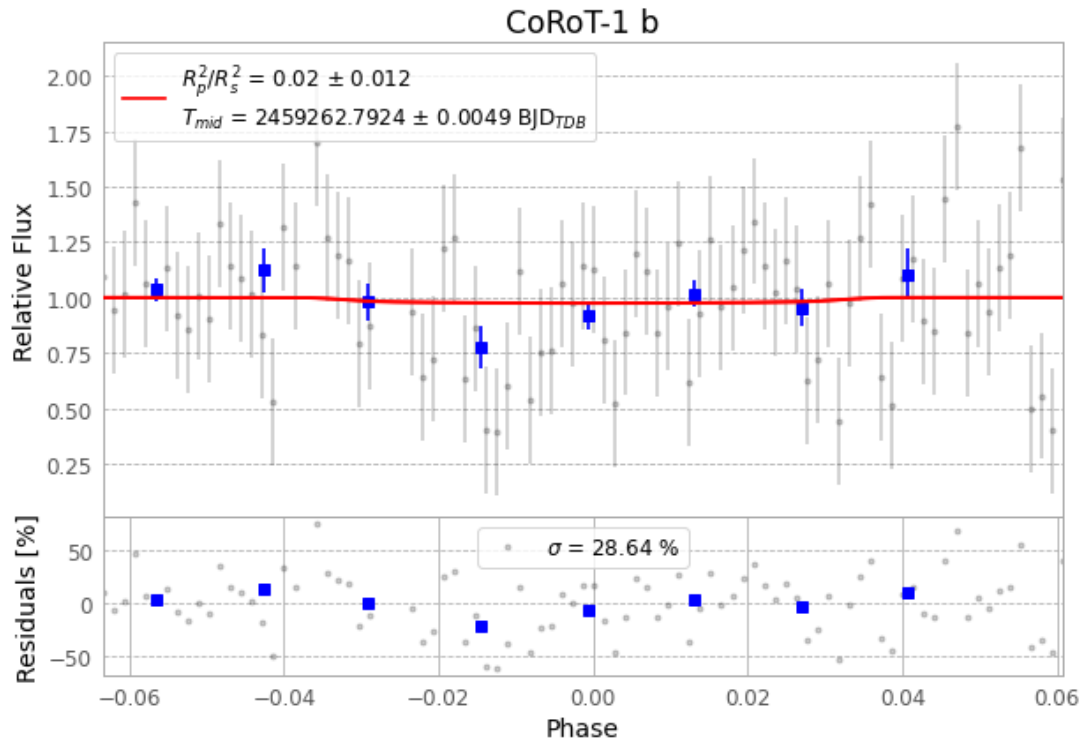


Figure 1 — FinalLightCurve_CoRoT-1 b_2021-02-16.png (SHA-256 33b28043921a5451...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 90], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08aea17b5d998c2b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

91 FITS frames (60 MB), CoRoT-1210217043613.FITS → CoRoT-1210217090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-210217.log (d680e13ca2af...), FinalLightCurve_CoRoT-1 b_2021-02-16.png (33b28043921a...), FinalLightCurve_CoRoT-1 b_2021-02-16.csv (df45036c5556...)

Compute resources

Wall time 133.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:53Z.